

WS22 - Understanding clValid's stability output

Study the output below and answer the following questions:

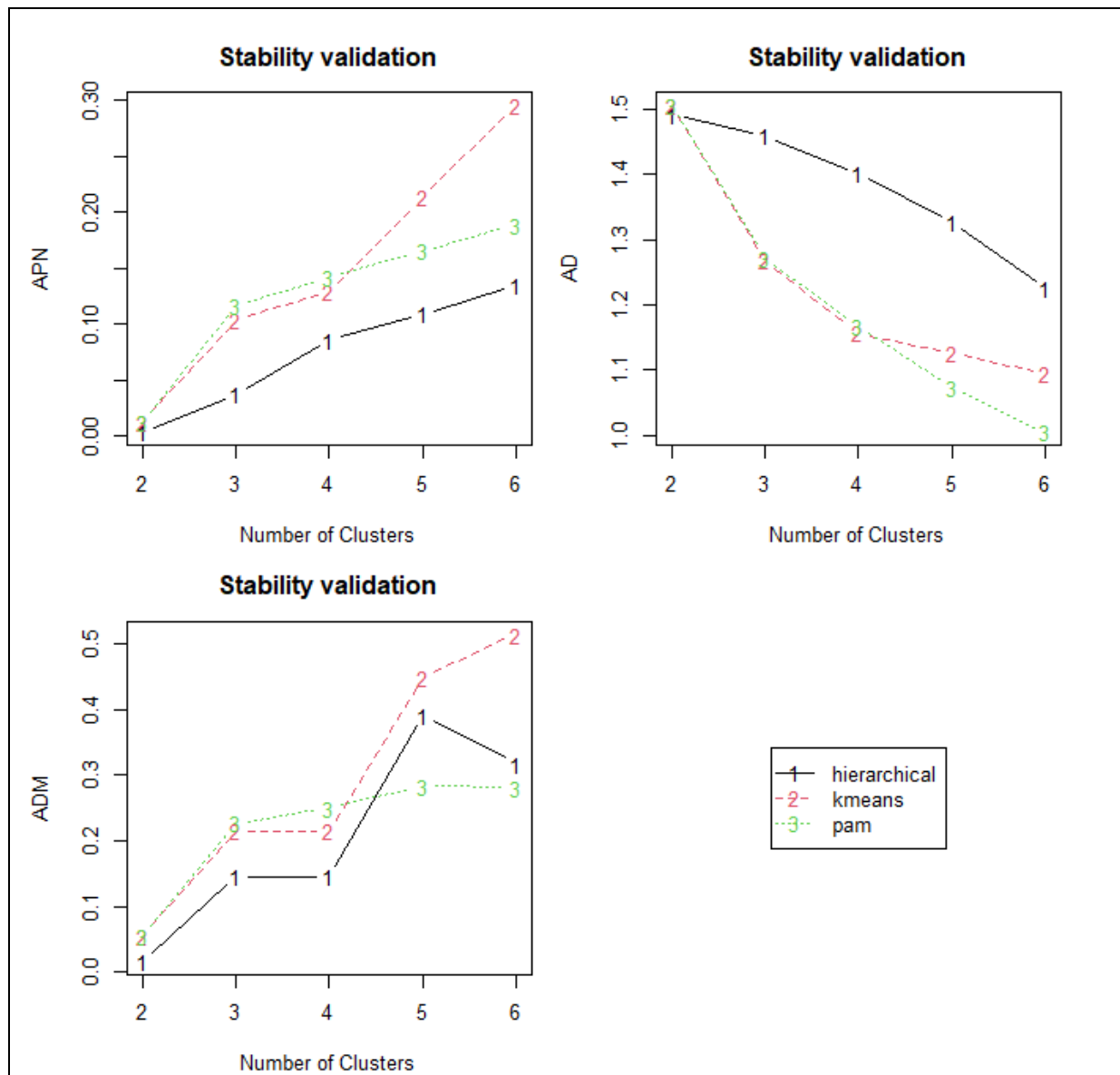
```
> library(clValid)
> df <- scale(iris[, -5])
> clmethods <- c("hierarchical", "kmeans", "pam")
> stab <- clValid(df, nClust = 2:6, clMethods = clmethods,
+               validation = "stability")
Warning message:
In clValid(df, nClust = 2:6, clMethods = clmethods, validation = "stability") :
  rownames for data not specified, using 1:nrow(data)
> # Display only optimal Scores
> optimalScores(stab)
```

	Score	Method	Clusters
APN	0.003266667	hierarchical	2
AD	1.004288856	pam	6
ADM	0.016087089	hierarchical	2
FOM	0.455750052	pam	6

- (a) For APM, AD, ADM and FOM, “stable” clusters should have low or high scores? Look at the plots in Page 2 – confirm that your answer is correct by checking if clValid package is reporting low or high scores (APM, AD, ADM, FOM – the plot for FOM is not included) as most “stable” method.

- (b) The output from the first page only provides the “best” stability scores without any info on which method (and value of k) is second best. To visualize stability scores (APN, AD, ADM, FOM), you should make use of `clValid`’s documentation (page 15) and produce the following plots. FOM was not included here but for your project, you are expected to include FOM as well (which requires making some modifications to the code below).

```
> par(mfrow=c(2,2),mar=c(4,4,3,1))
> plot(stab, measure=c("APN", "AD", "ADM"), legend=FALSE)
> plot(nClusters(stab), measures(stab, "APN")[,1], type="n", axes=F, xlab="", ylab="")
> legend("center", clusterMethods(stab), col=1:9, lty=1:9, pch=paste(1:9))
```



Answer the following:

Based on APN, the best method is _____ with k = _____

The second best method is _____ with k = _____

Based on AD, the best method is _____ with k = _____

The second best method is _____ with k = _____

Based on ADM, the best method is _____ with k = _____

The second best method is _____ with k = _____

2024 Update: Although the graphs on page 2 are very useful, it is best to accompany these graphs with actual values:

```
> summary(stab)

Clustering Methods:
  hierarchical kmeans pam

Cluster sizes:
  2 3 4 5 6

Validation Measures:

                2         3         4         5         6

hierarchical APN  0.0033  0.0370  0.0859  0.1088  0.1342
              AD   1.4924  1.4601  1.4015  1.3269  1.2251
              ADM  0.0161  0.1451  0.1448  0.3919  0.3176
              FOM  0.5957  0.5809  0.5571  0.5334  0.5101
kmeans       APN  0.0128  0.1034  0.1290  0.2130  0.2950
              AD   1.5060  1.2685  1.1568  1.1269  1.0949
              ADM  0.0555  0.2152  0.2147  0.4499  0.5147
              FOM  0.6049  0.5206  0.4888  0.4805  0.4910
pam          APN  0.0128  0.1162  0.1420  0.1655  0.1886
              AD   1.5060  1.2721  1.1665  1.0726  1.0043
              ADM  0.0555  0.2266  0.2500  0.2834  0.2814
              FOM  0.6049  0.5032  0.4828  0.4789  0.4558

Optimal Scores:

      Score Method      Clusters
APN 0.0033 hierarchical 2
AD  1.0043 pam          6
ADM 0.0161 hierarchical 2
FOM 0.4558 pam          6
```